

Théo Sourget

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EDUCATION

Université de Rouen

Master in Data Science and Engineering, highest honors, Valedictorian

Rouen, France
2021-2023

Université de Rouen

Bachelor in Computer Science and Data Science, honors, Valedictorian

Rouen, France
2020-2021

IUT de Vannes

2 Years University Diploma in Computer Science (DUT Informatique)

Vannes, France
2018-2020

CURRENT & RECENT EMPLOYMENT

IT University of Copenhagen

PhD Student, Pattern Recognition Revisited lab

Copenhagen, Denmark
September 2025 – Current

- I study data quality and the evaluation of machine learning algorithms
- Supervised by Veronika Cheplygina

CentraleSupélec, University of Paris-Saclay

Research Engineer, MICS laboratory

Gif-sur-Yvette, France
April 2025 – September 2025

- I studied CLIP-based models on the classification of chest X-rays
- I assessed their fairness across subgroups of patients and their robustness to shortcut learning

IT University of Copenhagen

Research Assistant, Pattern Recognition Revisited lab

Copenhagen, Denmark
October 2023 – January 2025

- I assessed the usage of non-relevant information in medical images (chest x-rays, eye fundus) by CNN models
- I studied the presence of public medical datasets in scientific papers and created open-access tools based on the research
- I defined and wrote a PhD funding application on the evaluation of multi-modal datasets and models

IT University of Copenhagen

Assistant Lecturer, Department of Computer Science

Copenhagen, Denmark
October 2023 – July 2024

- I assisted students during practical sessions of the Data In the Wild Course of Fall 2023 (data processing, visualisation, web scraping)
- I prepared and gave lectures to bachelor's students of the Project in Data Science course of Spring 2024 (Git, decision trees, ensemble of classifiers, common problems in machine learning projects)

Capgemini Engineering

Data Scientist Intern, Medic@

Illkirch-Graffenstaden, France
April 2023 – September 2023

- I performed teeth detection, segmentation and numbering in X-ray dental panoramic to help diagnose rare diseases for both adults and children.
- I compared the performances of Mask-RCNN and Detection Transformer (DETR)
- I defined an algorithm to generate new dental panoramics using existing images and inpainting models
- I used different generative models (GAN, diffusion models) to generate new teeth

SELECTED PUBLICATIONS

- Dataset Diversity Metrics and Impact on Classification Models,
T. Sourget, N. Claßen, J.J. Xu, R. van der Goot, V. Cheplygina, BISCUIT ECCV Workshop, 2026
- Look What the Probes Dragged In! Real-World Chest X-ray Shortcuts in MedCLIP,
N. Pedersen, R. Sydendal, V. Cheplygina, **T. Sourget**, FAIMI MICCAI Workshop, 2026
- Fairness and Robustness of CLIP-Based Models for Chest X-rays,
T. Sourget, D. Restrepo, C. Hudelot, E. Ferrante, S. Christodoulidis, M. Vakalopoulou, FAIMI MICCAI Workshop, 2025
- Mask of Truth: Model Sensitivity to Unexpected Regions of Medical Images,
T. Sourget, M. Hestbek-Møller, A. Jiménez-Sánchez, J.J. Xu, V. Cheplygina. Journal of Imaging Informatics in Medicine, 2025

- In the Picture: Medical Imaging Datasets, Artifacts, and their Living Review, Amelia Jiménez-Sánchez, [...], **Théo Sourget**, [...], Veronika Cheplygina, ACM FAccT, 2025
- Copycats: the many lives of a publicly available medical imaging dataset, A. Jiménez-Sánchez, N.R. Avlona, D. Juodelyte, **T. Sourget**, C. Vang-Larsen, A. Rogers, H.D. Dariusz Zajac, V. Cheplygina. NeurIPS Datasets and Benchmarks Track, 2024.
- [Citation Needed] Data usage and citation practices in medical imaging conferences. **T. Sourget**, A. Akkoç, S. Winther, C. L. Galsgaard, A. Jiménez-Sánchez, D. Juodelyte, C. Petitjean, and V. Cheplygina. Medical Imaging with Deep Learning (MIDL), 2024.
- Can Segformer be a true competitor to U-net for medical image segmentation? **T. Sourget**, S. N. Hasany, F. Mériaudeau, and C. Petitjean. Medical Image Understanding and Analysis (MIUA), 2023.

OTHER ACHIEVEMENTS

Presentations

- **BISCUIT ECCV workshop 2026 (Oral and Poster):** Dataset Diversity Metrics and Impact on Classification Models
- **FAIMI MICCAI workshop 2025 (Poster):** Fairness and Robustness of CLIP-Based Models for Chest X-rays
- **MICS Seminar (Talk):** Practices and problems with publicly available medical datasets
- **D3A Conference 2.0 2024 (Poster):** Mask of truth: model sensitivity to unexpected regions of medical images
- **"In the Picture: Medical Imaging Datasets" workshop (Oral):** [Citation Needed] Data usage and citation practices in medical imaging conferences
- **MIDL 2024 (Oral and Poster):** [Citation Needed] Data usage and citation practices in medical imaging conferences
- **D3A Conference 1.0 2024 (Poster):** [Citation Needed] Data usage and citation practices in medical imaging conferences
- **GRETSI 2023 (Poster):** Can Segformer be a true competitor to U-net for medical image segmentation?

Fundings and awards

- **Best Paper Award at BISCUIT ECCV workshop 2026** for Dataset Diversity Metrics and Impact on Classification Models
- **Best Poster Presentation Award at FAIMI MICCAI workshop 2025** for Fairness and Robustness of CLIP-Based Models for Chest X-rays
- **DDSA visit grant** of EUR 1645 in 2023 to work with Dr Veronika Cheplygina at the IT University of Copenhagen
- **Graduate school MinMacs scholarship** of EUR 15 000 (funded by the Normandy region) during both the first and second year of my Master's degree, giving access to funding, a supervisor and an immersion in laboratories

Other roles

- Member of the organisation committee of the "Datasets through the Looking-Glass" webinars
- Assisted in the technical setup of the "In the Picture: Medical Imaging Datasets" workshop

REVIEWS

- BISCUIT ECCV workshop 2026
- PLOS Digital Health journal
- IEEE Transactions on Medical Imaging journal
- FAIMI MICCAI workshop 2025, 2026
- MELBA Special issue on Fairness of AI in Medical Imaging
- Medical Imaging with Deep Learning (MIDL) conference 2025

SUPERVISION

- Alexander Winther Nielsen - Target-model Agnostic Evaluation Compression for image-classification Models (MSc thesis, Fall 2026)
- Nikolette Zoe Pedersen & Regitze Julie Sydendal - Investigating Real-World Shortcuts in MedCLIP for Chest X-ray Images (MSc thesis, Spring 2026)
- Nikolette Zoe Pedersen & Regitze Julie Sydendal - Impact of demographic data diversity on model training dynamics using data maps (MSc research project, Fall 2025)

TEACHING

- ITU, Project in Data Science - BSc level (lecturer) Spring 2024, 2026
- ITU, Data in the wild - MSc level (teaching assistant & lecturer) Fall 2023, 2024, 2025